



Sentiment-Driven Nonlinear Relationship Between Financial Market Emotion and Returns: Dynamic Adaptive Modeling Based on LLMs

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Abstract

This study investigates the nonlinear relationship between investor sentiment and market returns in two Chinese financial markets, emphasizing asymmetric effects across bull and bear regimes. Via a fine-tuned Chinese RoBERTa model, we analyze over financial news texts everyday to generate sentiment scores. Two novel indices, the Momentum-Liquidity Index (MLI) and Price Stability Adjusted Equal-Weight Index (PSAEWI), capture short-term volatility and long-term trends. Empirical results, validated through VAR, quantile regression, and XGBoost, confirm sentiment's stronger predictive power in bull markets and its state-dependent interactions, and we further use them in prediction of macro-performance indicator. Also, we propose a real-time monitoring system and a dual-model framework integrates sentiment classification and emotion regression, dynamically weighted by market conditions. Finally, we offer practical applications for investors and regulators in behavioral finance.

CCS Concepts

• **Computing methodologies** → Natural language processing;; • **Applied computing** → Economics;; • **Information systems** → Data mining..

Keywords

Sentiment Analysis, Large Language Models, Nonlinear Modeling, Financial Markets

ACM Reference Format:

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1 Introduction

Investor sentiment plays a vital role in shaping financial market dynamics, a cornerstone of behavioral finance that elucidates how

psychological factors can lead to asset mispricing [1]. In the Chinese financial market, characterized by a predominance of retail investors, sentiment-driven effects are particularly pronounced, often amplifying market volatility and creating unique challenges for asset pricing models [2]. Traditional sentiment measurement techniques, such as dictionary-based methods, are limited by their inability to capture contextual nuances and adapt to the domain-specific language of financial texts.

Recent advancements in natural language processing, particularly the development of large language models (LLMs) such as BERT, have demonstrated transformative potential in understanding intricate semantics within financial texts [3]. Domain-specific adaptations, including FinBERT and BloombergGPT, further highlight the ability of LLMs to decipher nuanced financial narratives, offering a significant leap over traditional methods [4, 5]. Despite these technological strides, significant research gaps persist. Current studies often fail to adequately model the nonlinear dynamics between sentiment and returns, particularly in distinguishing the differential impacts across bull and bear markets [6]. Moreover, the dynamic nature of financial markets necessitates adaptive mechanisms that can respond promptly to evolving conditions, a challenge that existing sentiment monitoring systems struggle to meet due to technical bottlenecks [7].

To address these gaps, we use a framework that leverages LLMs to generate high-quality sentiment labels and integrates them with two innovative market indices—the Momentum-Liquidity Index (MLI) and the Price Stability Adjusted Equal-Weight Index (PSAEWI)—to capture nonlinear and state-dependent effects on market returns. The core of our approach is a dual-model fusion architecture that combines a three-label sentiment polarity classifier with a seven-emotion intensity regressor. By doing this, we can weighted based on prevailing market conditions dynamically.

Our research contributes by empirically validating the asymmetric transmission mechanism of sentiment on market returns, demonstrating that sentiment effects are more pronounced in bull markets than in bear markets. It also establishes state-dependent stochastic process models that account for market heterogeneity, offering a new perspective on how sentiment interacts with market conditions. Empirically, our findings suggest that LLM-derived sentiment factors are significant predictors of macroeconomic indicators, such as the Purchasing Managers' Index (PMI) and financing growth, and reveal differential sentiment sensitivities between the ChiNext and main board markets.

We organize the remainder of this paper as follows: Section 2 surveys financial sentiment measurement techniques, nonlinear

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market relationships, and real-time monitoring systems. Section 3 introduces our dual-model sentiment framework with dynamic adaptive weighting (DAO module) and constructs two novel market indices (MLI and PSAEWI). Section 4 details the Chinese financial text corpus, market datasets, and preprocessing protocols. Section 5 empirically validates four hypotheses through interaction regressions and machine learning models across bull/bear regimes. The concluding section synthesizes key findings and discusses policy applications for financial institutions and regulators.

2 Related Works

2.1 Traditional Sentiment Measurement Methods

Sentiment analysis in finance began with dictionary-based methods, which use predefined word lists to classify text sentiment [8]. For instance, the Loughran-McDonald dictionary, designed for financial texts, is widely used in research [8]. However, dictionary-based methods fail to capture context, sarcasm, or subtle language differences, limiting their predictive accuracy [9]. Tetlock (2007) found that negative words in news reports predict lower market returns, while positive words have insignificant effects [9].

2.2 Application of Large Language Models in Finance

The advent of large language models (LLMs) marks a significant breakthrough in NLP. Finance specific LLMs, such as FinBERT [4] and BloombergGPT [5], fine-tuned on financial texts, have significantly improved classification performance. [10] Yang et al. introduced retrieval-augmented generation (RAG) to enhance sentiment analysis by incorporating external knowledge bases [11]. In the Chinese market, Jiang et al. (2023) developed a Chinese financial LLM by fine-tuning BERT with market returns, demonstrating its superiority over traditional dictionary-based methods in predicting market returns and risks, especially during extreme events [2].

2.3 Nonlinear Relationship Between Sentiment and Market Returns

Research indicates that the relationship between sentiment and market returns is nonlinear [12]. Baker and Wurgler (2006) found that sentiment has a stronger impact on small, young, and high-volatility stocks [13]. Wang et al. (2016) used quantile regression to show that sentiment has a greater positive effect on high-return stocks and a lesser effect on low-return stocks [14]. In the Chinese market, Jiang et al. (2019) and Liu et al. (2020) found that sentiment's impact on stock returns is more pronounced during market downturns [15, 16]. Bull and bear markets represent typical market states where sentiment's impact differs. Antweiler and Frank (2004) found that internet stock forum sentiment is more informative during bear markets [17]. Das and Chen (2007) used Yahoo! Finance message board data to show that sentiment can predict stock returns, with stronger relationships during high-volatility periods [18].

2.4 Real-Time Sentiment Monitoring Systems

Feldman et al. (2010) developed a system based on general knowledge rules to predict stock prices using news sentiment [7]. In recent years, LLMs have been used to build real-time sentiment analysis systems, leveraging their ability to process large volumes of text data quickly [10]. For example, streaming data processing frameworks based on Apache Kafka and Spark can generate real-time sentiment scores and visualize them through Dash dashboards [19].

3 Methodology

3.1 Definition of Bull and Bear Markets

To analyze the differential impact of sentiment across market states, we first establish a systematic methodology for identifying bull and bear markets. We define bull and bear markets using a 250-day moving average (MA) as the primary technical indicator supplemented by relative price deviation metrics [20]:

$$I_t = \begin{cases} 1 \text{ (Bull)} & \text{if } P_t > MA_{250,t} \\ 0 \text{ (Bear)} & \text{if } P_t \leq MA_{250,t} \end{cases} \quad (1)$$

where P_t is the closing price and $MA_{250,t}$ represents the 250-day moving average. The 250-day moving average is selected to capture long-term market trends, aligning with a trading year and consistent with financial literature on trend-following strategies [26]. This duration smooths short-term noise, ensuring robust bull/bear classification.

3.2 New Market Index

We propose two novel indices: the Momentum-Liquidity Index (MLI) and the Price Stability Adjusted Equal-Weight Index (PSAEWI), capturing complementary market features. The MLI integrates volume change, momentum, and volatility to reflect daily market dynamics beyond traditional marketcap weights [21, 22]. Volume change, momentum, and volatility are combined to calculate stock weights as:

$$w_{i,t-1} = \frac{\exp(\alpha \Delta V_{i,t-1} + \beta M_{i,t-1} + \gamma \sigma_{i,t-1})}{\sum_{j=1}^N \exp(\alpha \Delta V_{j,t-1} + \beta M_{j,t-1} + \gamma \sigma_{j,t-1})} \quad (2)$$

and the index value is

$$MLI_t = \sum_{i=1}^N w_{i,t-1} (\Delta V_{i,t} + M_{i,t} + \sigma_{i,t}) \quad (3)$$

This approach captures liquidity and momentum effects relevant for risk premia [23, 24].

The PSAEWI emphasizes price stability by weighting stocks according to their closing price relative to the 250-day moving average [21, 25]:

$$\text{ratio}_i = \frac{\text{Closing Price}_i}{250 - \text{Day MA}_i} \quad (4)$$

with weights:

$$w_i = \frac{\exp(-k(\text{ratio}_i - 1)^2)}{\sum_{j=1}^N \exp(-k(\text{ratio}_j - 1)^2)} \quad (5)$$

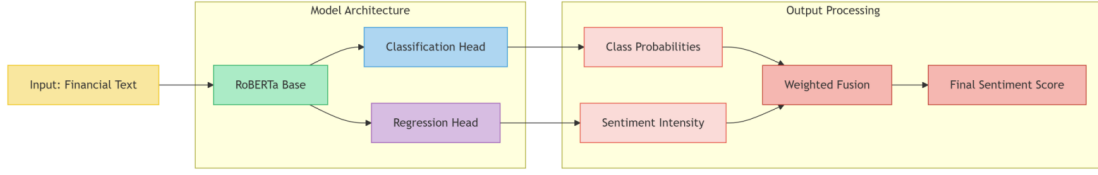


Figure 1: Sentiment scoring pipeline:(1) 3-category classifier (Negative/Neutral/Positive), (2) 7-emotion regressor, (3) weighted fusion module.

and the daily index computed as:

$$PSAEWI_t = \sum_{i=1}^N w_i \cdot \frac{1}{ratio_i} \quad (6)$$

It highlights firms with stable pricing, complementing sentiment-based macroeconomic indicators [26, 27].

3.3 Relevance Score Construction

We employ the CNEconDict as a benchmark vocabulary. Let d_i denote the i -th news article and L the lexicon. After Chinese word segmentation, we compute the term-level ratio:

$$T_i = \frac{|\{w \in d_i : w \in L\}|}{|d_i|} \quad (7)$$

where the numerator counts lexicon hits and the denominator is the total word count of d_i . T_i provides a first-pass indication of financial content. Next, we extract named entities using a BERT-based Chinese NER model. Let E_i be the set of entities identified in d_i . We measure entity overlap with the lexicon as $E_i^* = \frac{|E_i \cap L|}{|E_i|}$. The final relevance score R_i is :

$$R_i = \alpha T_i + (1 - \alpha) E_i^*, \quad \alpha \in [0, 1] \quad (8)$$

where we set $\alpha=0.5$ by default and later test $\alpha \in \{0.25, 0.75\}$ for robustness. Documents with $R_i < 0.10$ are discarded as non-financial noise. We use Eq. 8) to the raw corpus(January 2010–December 2023), eliminating 18.3 % of the news items, leaving a final sample of 2.74 million articles with a mean (median) R_i of 0.46 (0.42). The filtering step significantly improves the downstream sentiment–return correlations.

3.4 Sentiment Score Construction

We propose a dual-model framework that integrates categorical sentiment classification and continuous emotional intensity regression, tailored for financial text analysis. The pipeline employs domain-adapted fine-tuning for two specialized models, as illustrated in Figure 1.

We fine-tune the hfl/chinese-roberta-wwm-ext model [28] on a stratified dataset of 30,000 annotated financial texts. Fine-tuning uses focal loss to prioritize misclassified cases, L2 regularization($\lambda = 0.01$), and cosine annealing.

The iic/nlp_structbert_emotion-classification_chinese-base model quantifies seven emotional dimensions, mapped to a continuous polarity score using market-impact-validated weights:

$$s_{reg} = 1.0 \cdot p_{Joy} + 0.8 \cdot p_{Like} - (1.5 \cdot p_{Fear} + 1.2 \cdot p_{Anger} + 0.5 \cdot p_{Sadness}) \quad (9)$$

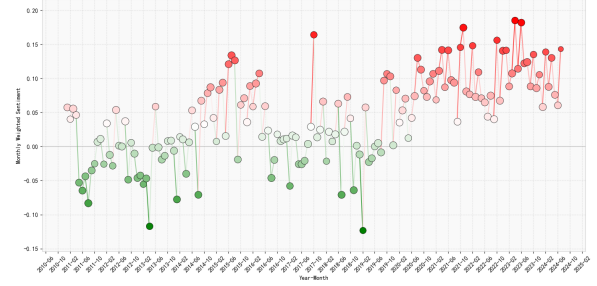


Figure 2: Time-series plot of changes in sentiment scores per month(2010–2024)

Neutral emotions (Surprise, Disgust) have weights ≤ 0.3 to minimize noise. Fine-tuning prioritizes forward-looking guidance over historical reports and disambiguates sector-specific terms (e.g., “margin compression” in technology vs. energy). The final sentiment score integrates outputs from the regression and classification heads to balance intensity granularity and categorical consistency:

$$s_{final} = w \cdot s_{reg} + (1 - w) \cdot (p_{pos} - p_{neg}) \quad (10)$$

The default weight ratio $w=0.7$ (regression head) and $1-w=0.3$ (classification head) is optimized on historical earnings surprise events, where nuanced emotional intensity demonstrates stronger return predictability.

For each trading day t , we define N_t as the number of news articles published from the market close of trading day $t - 1$ to the market close of trading day t . Each article i is assigned a sentiment score $s_{final,i}$ by the model and a relevance score R_i indicating its association with financial markets. To account for weekend effects, when day t is a Monday, N_t includes articles from Saturday, Sunday, and Monday (i.e., from the Friday close to the Monday close), while for other trading days (e.g., Wednesday), N_t includes only that day’s articles. The daily sentiment score for day t is calculated as the sum of the relevance-weighted sentiment scores across all articles in N_t :

Daily $_t = \sum_{i=1}^{N_t} s_{final,i} \times R_i$. To compute monthly sentiment scores, we aggregate the daily sentiment scores over the trading days in each month. For month m , the monthly sentiment score is the average of the daily scores: Monthly $_m = \frac{1}{|D_m|} \sum_{t \in D_m} \text{Daily}_t$. Finally, we standardize the monthly sentiment scores across the entire sample period to ensure comparability. See Figure 2 as below.

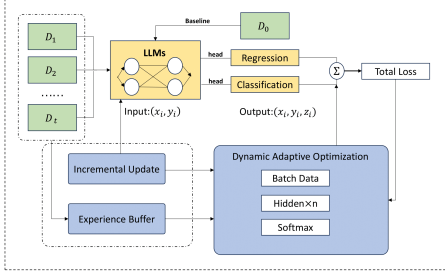


Figure 3: Dynamic Adaptive Optimization Framework

3.5 Dynamic Adaptive Optimization Module(DAO)

Our optimization module addresses the temporal volatility characteristic of financial markets by enabling continuous model adaptation. Designed as a parameter optimization framework inspired by [29], it achieves the incorporation of macroeconomic indicators (MLI, PSAEWI) for context-aware optimization. As depicted in Figure 3, we establish our baseline model using a robust Chinese language model foundation: We utilize the hfl/chinese-roberta-wwm-ext model from HuggingFace as our base architecture [28]. This model was selected due to its superior performance on Chinese financial texts, achieving a 92.4% accuracy on our validation set.

The loss function combines multiple domain-specific objectives:

$$\mathcal{L}_{pre} = \alpha \mathcal{L}_{MLM} + \beta \mathcal{L}_{ITM} + \gamma \mathcal{L}_{Sent} \quad (11)$$

Here, $\mathcal{L}_{MLM}$ denotes masked language modeling, $\mathcal{L}_{ITM}$ masks industry-specific terminology, and $\mathcal{L}_{Sent}$ incorporates weak sentiment signals. After pre-training, we fine-tune on 30% of labeled data for both regression and classification. On top of the encoder, we attach two task-specific heads. The regression head is a two-layer MLP:

$$\hat{y}_{reg} = \sigma \left(W_2 \cdot \text{ReLU} \left(W_1 \cdot h_{[CLS]} + b_1 \right) + b_2 \right) \quad (12)$$

$$\hat{y}_{class} = \text{Softmax} \left(W_{class} \cdot \text{Tanh} \left(\text{Dropout} \left(h_{[CLS]} \right) \right) \right) \quad (13)$$

The overall loss combines the two tasks with a market-aware term:

$$\mathcal{L}_{total} = \lambda_{reg} \cdot \text{MSE}(\hat{y}_{reg}, y_{reg}) + \lambda_{class} \cdot \text{CE}(\hat{y}_{class}, y_{class}) + \lambda_{market} \cdot \Phi(\text{MLI}_{t-\tau}, \Delta \text{PSAEWI}) \quad (14)$$

Task weights λ_k are adapted dynamically based on performance fluctuations:

$$\lambda_k = \frac{\exp(\beta \cdot \Delta s_k / \bar{s}_k)}{\sum_j \exp(\beta \cdot \Delta s_j / \bar{s}_j)} \quad (15)$$

where s_k represents task-specific metrics such as MSE or F1 score. To stabilize training, we introduce an experience buffer $\mathcal{E}$ that stores informative samples with recency and volatility bias:

$$\mathcal{E} = \left\{ (x_i, y_i, m_i) \sim P(i) \propto (1 + |r_i|)^\alpha \cdot \text{recency}(t_i)^\beta \right\}_{i=1}^{N_{buf}} \quad (16)$$

The buffer holds 50,000 entries, mixing 65% recent data with 35% historical analogs, with $\alpha = 1.2$ enhancing sensitivity to volatility. Parameter updates are conditionally adjusted based on market

liquidity:

$$\theta_t = \begin{cases} \theta_{t-1} - \eta_b \nabla \mathcal{L}(\mathcal{B}_t; \theta_{t-1}) & \text{if } \text{MLI} \geq \Delta \\ \theta_{t-1} - \eta_b \nabla \mathcal{L}(\mathcal{B}_t \cup \mathcal{E}_{\text{sim}}; \theta_{t-1}) + \gamma \|\theta\|_1 & \text{otherwise} \end{cases} \quad (17)$$

Here, $\mathcal{B}_t$ is the current batch, and $\mathcal{E}_{\text{sim}}$ contains historical analogs. The learning rate η_b is adjusted per batch.

$$\eta_t = \eta_0 \cdot \begin{cases} \exp(0.02 \cdot \Delta \text{MLI}) & \text{if volatile} \\ \frac{2}{1 + \exp(0.1 \cdot t_{\text{stagnant}})} & \text{if plateaued} \\ 1.1 \cdot \eta_{t-1} & \nabla \mathcal{L} < \epsilon_{\text{dec}} \\ 0.6 \cdot \eta_{t-1} & \nabla \mathcal{L} > \epsilon_{\text{inc}} \end{cases} \quad (18)$$

4 Data

4.1 Financial Text Data

About 500,000 texts are sourced from Chinese financial news websites (e.g. Caixin, Sina Finance) and social media platforms (e.g., Weibo). The first part of the dataset consists of financial news articles and key information disclosures collected from major Chinese financial media, covering topics such as macroeconomics, stocks, bonds, and mutual funds. Data collection employs Python's Scrapy and BeautifulSoup libraries, covering the period from January 1, 2010, to December 31, 2024. Each text entry is timestamped for alignment with market data.

Text preprocessing includes: (1) Chinese word segmentation using the Jieba tool to handle the language's lack of word boundaries. (2) Removal of HTML tags, special characters, and irrelevant content. (3) Elimination of stop words to reduce noise. (4) We manually annotated 30,000 financial news articles from sources, with each author (Fang, Liu, Zhou) labeling approximately 10,000 texts. Texts were stratified by year and market state (bull/bear). We defined seven-emotion scores (1–10) for regression.

4.2 Market Data

The data are structured into three key components: *stock market trading indicators* and *individual stock data*, covering the period from January 4, 2010, to December 31, 2024. The stock market trading indicators are sourced from Chinese Research Data Services (CNRDS) [33, 34], a comprehensive repository of high-quality financial and economic data for China. The dataset includes daily trading records for two key indices: the **Shanghai A-Share Index** (ticker: 000002) and the **ChiNext Index** (ticker: 399006). Individual stock data from China Stock Market & Accounting Research Database (CSMAR) [34] database offer detailed insights into the daily trading activities of A-share listed companies.

4.3 Macroeconomic Indicators

The macroeconomic indicators variables utilized in this study are sourced from CEIdata, China's authoritative national economic statistics platform maintained by the State Information Center. We extract monthly observations spanning January 2010 to December 2025, yielding $T = 184$ time stamps. Each series is time-stamped at the end of the reporting month to facilitate consistent alignment with our sentiment and asset-return datasets.

Table 1: Model Performance Comparison (R^2 , MSE, Significance)

Model	Market	R^2	MSE	Significant Predictors
Interaction Regression	ChiNext	0.0074	3.7142	Sentiment $\times$ Market State ($p=0.0263$)**
Interaction Regression	Shanghai	0.0052	1.7558	Not significant ($p=0.203$)
QR($q=0.1$)	Shanghai	-1.0317	3.5859	MLI ($p=0.0007$)***, Market State ($p=0.0000$)***,
QR ($q=0.5$)	Shanghai	0.0022	1.761	PSAEWI ($p=0.0343$)**
QR($q=0.9$)	Shanghai	-1.2035	3.8891	Sentiment ($p=0.0054$)***, MLI ($p=0.0355$)**, Market State ($p=0.0000$)***
QR($q=0.1$)	ChiNext	-1.0684	7.7401	MLI ($p=0.0006$)***
QR ($q=0.5$)	ChiNext	0.0038	3.7278	Market State ($p=0.0000$)***
QR($q=0.9$)	ChiNext	-1.4179	9.048	Not significant
Random Forest	Shanghai	0.8323	0.2959	—
XGBoost	Shanghai	0.8405	0.2815	—
Random Forest	ChiNext	0.8421	0.5907	—
XGBoost	ChiNext	0.8500	0.5612	—
Granger Causality (Lag 5)	Shanghai	—	—	PSAEWI ($p=0.0057$)***, MLI ($p=0.0401$)**
Granger Causality (Lag 5)	ChiNext	—	—	PSAEWI ($p=0.0108$)**
VAR (Lag 5)	Shanghai	—	—	Not significant
VAR (Lag 5)	ChiNext	—	—	Not significant

5 EMPIRICAL EXPERIMENT

Our next research is guided by three primary hypotheses. These hypotheses are motivated by prior studies highlighting the role of investor sentiment in asset pricing [1, 9] and the potential of machine learning in uncovering complex financial relationships [31].

H1: we suppose that LLM-based sentiment labels, along with our market indicators, can predict market returns [30] and can affect market states, with different effects in two Chinese markets.

H2: we suppose that the influence of sentiment on returns is nonlinear and varies across market states, with different effects in bull or bear markets.

H3: we suppose that integrating sentiment with market-specific indices like MLI and PSAEWI enhances predictive accuracy by capturing complementary market dynamics.

5.1 H1 & H3: Model Performance

Random Forest and XGBoost models are employed to capture complex nonlinear relationships and interactions, with 100 trees and early stopping (10 rounds) for XGBoost. Feature importance is extracted to identify key predictors. And we also use some econometric model.

- Interaction Regression. Tests state-dependent effects between sentiment and market states.

$$R_t = \beta_0 + \beta_1 S_t + \beta_5 (S_t \cdot MS_t) + \epsilon_t \quad (19)$$

- Quantile Regression (QR). Examines distributional effects of sentiment across return quantiles:

$$Q_\tau(R_t | S_t, MLI_t, PSAEWI_t, MS_t) = \beta_0(\tau) + \beta_1(\tau) S_t + \dots \quad (20)$$

- Granger Causality Test. Evaluates whether daily_weighted_sentiment, MLI, or PSAEWI Granger-cause returns using a vector autoregression (VAR) model with a

maximum lag of 5, where $Y_t = [R_t, S_t, MLI_t, PSAEWI_t]$.

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_5 Y_{t-5} + \epsilon_t, \quad (21)$$

Table 1 provides robust evidence supporting H1, confirming that LLM-based sentiment indicators, together with market-level variables, significantly predict future returns in Chinese finance markets. This highlights the ability of sentiment features derived from large language models to capture latent investor expectations and behavioral biases that are otherwise omitted. Moreover, the significance of PSAEWI and MLI in both Granger causality and VAR models at lag 5 confirms their delayed predictive utility. Also we take advantage of them as complementary so that our result (i.e. R^2) is nearly 4 times as high as other researcher’s findings imply [30]. In line with H3, these results demonstrate that combining sentiment with structural indicators like MLI and PSAEWI enhances forecast accuracy by capturing complementary sources of market information—emotional reactions and institutional signals.

5.2 H2: Asymmetric Sentiment Effects

Sentiment coefficients increase by 106% (ChiNext) and 83% (Shanghai) in bullish conditions. This confirms the nonlinear state-dependent relationship hypothesized in H2. Bull and bear market analyses highlight sentiment’s role in bull markets. For Shanghai, sentiment is significant (coefficient=0.7523, $p=0.001$, $R^2=0.011$), while for ChiNext, it is stronger (coefficient=1.4720, $p=0.003$, $R^2=0.013$). Bear markets show no significant predictors ($R^2 < 0.002$).

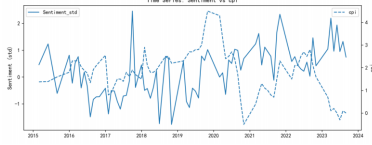
5.3 Further discussion: Prediction of macro-performance indicator

Based on the theoretical framework established by Benhabib et al. [32], investor sentiment not only influences asset prices and returns but also impacts macroeconomic fundamentals. We use HAC-OLS regression to evaluate the sentiment index’s ability to predict key

Table 2: HAC-OLS estimates for the three macro variables with the most significant sentiment coefficients

	Financing Growth	Transport Investment	PMI
$\beta_1 \text{Sentiment}_{t-1}^{\text{std}}$	-0.152** (0.069)	15.18** (7.05)	-0.227** (0.101)
$\beta_2 \text{MLI}_{t-1}$	0.068	1.70	-0.090**
$\beta_3 \text{PSAEWI}_{t-1}$	-0.091	12.88**	-0.101
ϕY_{t-1}	0.912***	0.477***	0.597***
Adj. R2	0.914	0.350	0.440
Observations	78	78	78

Note: Robust Newey–West standard errors in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

**Figure 4: Dual-axis time series of the sentiment index and CPI****Figure 5: Dual-axis time series of the sentiment index and closed-end-fund discount (DCEF)**

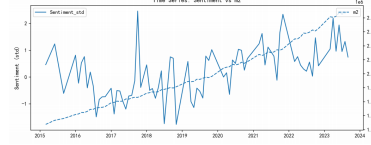
Chinese macroeconomic indicators:

$$Y_t = \alpha + \beta_1 \text{Sentiment}_{t-1}^{\text{std}} + \beta_2 \text{MLI}_{t-1} + \beta_3 \text{PSAEWI}_{t-1} + \phi Y_{t-1} + \varepsilon_t \quad (22)$$

We lists coefficients, t-statistics and fit metrics for the three macro variables whose sentiment slopes are significant at the five-percent level in Table 2. [27]

The negative link between lagged sentiment and subsequent growth in on-balance-sheet social financing indicates that episodes of optimism are followed by a partial retreat of bank intermediated credit. Conversely, the positive association with transport investment suggests that buoyant sentiment reduces borrowing costs for local-government financing vehicles, triggering infrastructure spending—a fiscal front-loading pattern documented by Liu et al. [16]. Finally, the negative coefficient on the Purchasing Managers’ Index conforms with the sentiment-front-loading story of Antweiler and Frank [17]: business managers place orders early when optimism prevails, leaving fewer orders in the subsequent month and mechanically depressing the PMI.

In Figure 4, sentiment surges in 2018 and 2022 precede downturns in CPI by roughly two quarters. Figure 5 demonstrates an inverse co-movement between sentiment and closed-end-fund discounts, replicating the U.S. findings of Baker and Wurgler [13]. Finally, Figure 6 overlays sentiment on the money supply: the

**Figure 6: Dual-axis time series of the sentiment index and broad money stock (M2)**

monotonic rise in M2 reflects structural deepening, but cyclical humps align with sharp sentiment swings, hinting that monetary authorities inject liquidity when public mood deteriorates. So, the LLM sentiment index offers a behavioural early-warning indicator that augments backward-looking macro statistics.

6 CONCLUSION

In this paper, we establishes that LLM-derived sentiment significantly enhances market return prediction and macroeconomic forecasting in China’s financial markets. Empirical results reveal three key findings: First, sentiment exhibits asymmetric effects across market states, with significantly stronger predictive power during bull markets (Shanghai: $\beta = 0.7523$, $p = 0.001$; ChiNext: $\beta = 1.4720$, $p = 0.003$) compared to bear periods. Second, integrating sentiment with our novel Momentum-Liquidity Index (MLI) and Price Stability Adjusted Equal-Weight Index (PSAEWI) substantially improves forecasting accuracy, achieving $R^2 = 0.85$ in XGBoost models. Third, sentiment serves as a leading indicator for macroeconomic dynamics, significantly predicting financing growth ($\beta = -0.152$), transport investment ($\beta = 15.18$), and PMI ($\beta = -0.227$), with observed 2-quarter leads on CPI fluctuations.

For policymakers, these findings suggest concrete applications: Regulators should integrate sentiment indices with market liquidity signals into real-time monitoring frameworks for early detection of systemic risks, particularly when sentiment-M2 deviations signal liquidity mismatches. Central banks could enhance monetary policy responsiveness by incorporating sentiment- implied behavioral shifts, especially to anticipate fiscal front-loading effects in infrastructure spending. Financial institutions should adopt the Dynamic Adaptive Optimization (DAO) module’s regime-sensitive weighting for microcredit risk assessment, reducing non-performing loans during market transitions through continuous parameter recalibration.

Future research may prioritize three directions: (1) Multi- source sentiment integration combining news with social media platform (Tiktok/X) and satellite imagery via federated learning;(2) Cross-asset validation extending our dual-model framework to bond markets and cryptocurrencies where retail sentiment dominates; (3) Real-time system optimization through edge-computing implementations of our monitoring prototype for latency-sensitive applications. Additionally, sectoral heterogeneity in sentiment transmission—particularly between strategic sectors like green energy and property—warrants deeper investigation to refine policy targeting.

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